Outline

- Statistical: Very Basic Concepts
- Probability: Very Basic Concepts

Statistical Experiment

- Statistical Experiment:
  - is a random or nondeterministic experiment.
  - cannot state the outcome.
  - can describe the set of all possible outcomes.
- If the experiment is repeated many times, a definite pattern or regularity can be detected.

It is easy to lie with statistics. It is hard to tell the truth without it.
- Andrejs Dunkels

Basic Definitions and Concepts

- A population (all objects of interest) is any specific collection of objects of interest.
- A sample (the objects examined) is any subset or subcollection of the population, including the case that the sample consists of the whole population, in which case it is termed a census.
- A measurement: A number or attribute computed for each member of a:
  - set of objects.
  - population or of a sample.
- The measurements of sample elements are collectively called the sample data (The measurements from a sample).

Basic Definitions and Concepts

- A parameter is a number that summarizes some aspect of the population as a whole.
- A statistic is a number computed from the sample data.
- Statistics is a collection of methods for collecting, displaying, analyzing, and drawing conclusions from data.
  - Descriptive statistics involves organizing, displaying, and describing data.
  - Inferential statistics involves drawing conclusions about a population based on information contained in a sample taken from that population
- Data:
  - Qualitative data are measurements for which there is no natural numerical scale, but which consist of attributes, labels, or other nonnumerical characteristics.
  - Quantitative data are numerical measurements that arise from a natural numerical scale.

Example

- Dark circles represents the elements of the population selected at random and that together form the sample.
- Each element of the sample there is a measurement of interest, denoted by (x, which have indexed as x_1 , … , x_n)
- These measurements collectively form the sample data set to calculate statistics.
- Such as compute the sample mean $\bar{x}$ and the sample proportion $\hat{p}$, approximations to the population mean (μ).

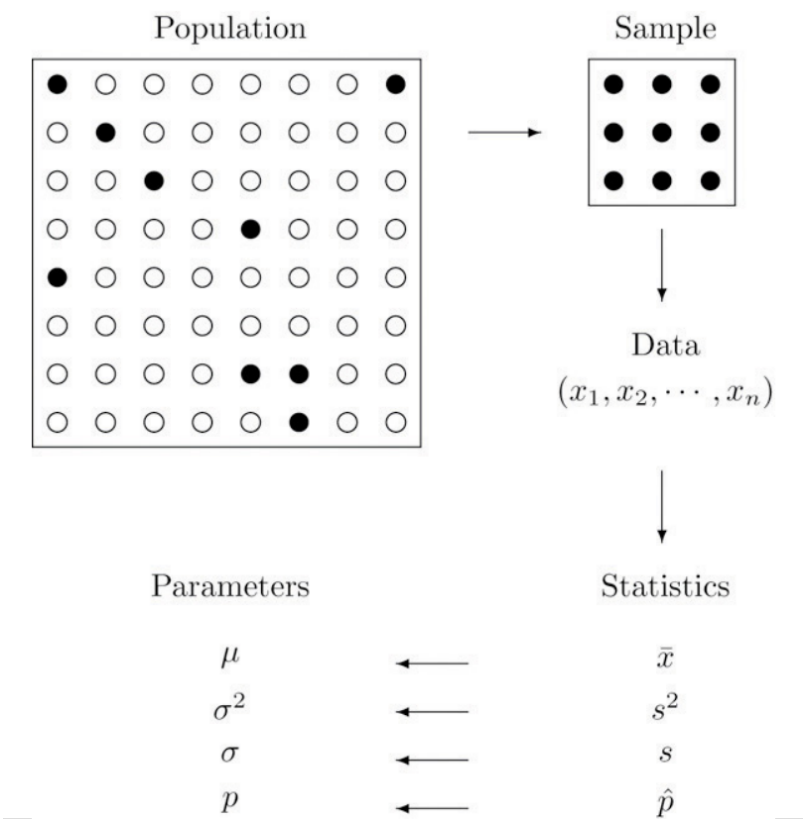
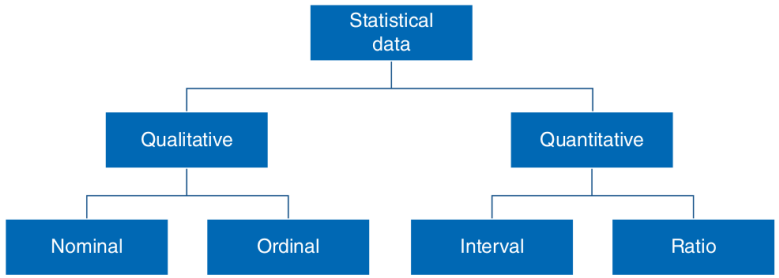


Image Src: Basic Definitions and Concepts

Data Classification

- Nominal data contain smallest amount of information. Symbols are used to label categories of a population.
- When the ordering of categories becomes important, the data collected are called ordinal. Examples: companies ranked according to the quality of their product.
- Interval data are numerical data, more informative than nominal and ordinal data but less informative than ratio data. Ex: temperature data.
- Ratio data (numerical) have the potential to produce the most meaningful information of all data types. Ex. All arithmetic operations.



Gupta, Bhisham C., and Irwin Guttman. Statistics and probability with applications for engineers and scientists. John Wiley & Sons, 2014.

Exercise on population or a sample

- Identify each of the following data sets as either a population or a sample:

1. The grade point averages (GPAs) of all students at KUC.
2. The GPAs of a randomly selected group of students at KUC Oslo college campus.
3. The ages of the nine Supreme Court Justices of Norway on January 1, 2020.
4. The gender of every second customer who enters a movie theater.
5. The lengths of Atlantic croakers caught on a fishing trip to the beach.

Solution on population or a sample

1. Population.
2. Sample.
3. Population.
4. Sample.
5. Sample.

Exercise on Quantitative or Qualitative

- Identify the following measures as either quantitative or qualitative:

1. The genders of the first 40 newborns in a hospital one year.
2. The natural hair color of 20 randomly selected fashion models.
3. The ages of 20 randomly selected fashion models.
4. The fuel economy in miles per gallon of 20 new cars purchased last month.
5. The political affiliation of 500 randomly selected voters.

Solution on Quantitative or Qualitative

1. Qualitative.
2. Qualitative.
3. Quantitative.
4. Quantitative.
5. Qualitative.

Presentation of Data

- There are two formats for presenting data sets:
  1. Data list: An explicit listing of all the individual measurements made on a sample.
  2. Data frequency table: A table listing each distinct value x and its frequency f.
- Example: Data obtained by measuring the age of 21 randomly selected students enrolled in freshman courses at KUC could be presented:
- Data list: 18 19 18 18 19 24 19 18 19 20 17

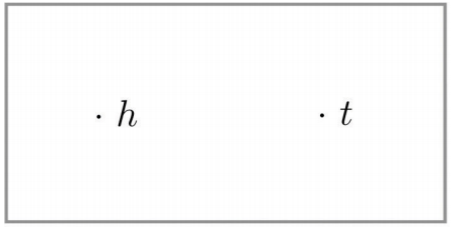
  18 18 22 21 20 20 18 17 18 19
- Data frequency table

x:	17	18	19	20	21	22	24
f:	2	8	5	3	1	1	1

Basic Concepts of Probability

- A random experiment is a mechanism that produces a definite outcome that cannot be predicted with certainty.
- The sample space associated with a random experiment is the set of all possible outcomes.
- An event is a subset of the sample space.
- An event E is said to occur on a particular trial of the experiment if the outcome observed is an element of the set E.
- Example: Construct a sample space for the experiment that consists of tossing a single coin.

  Solution: Let label the outcomes as "h" for heads and "t" for tails. Then the sample space is the set S = {h, t}.



Basic Definition

- The core of probability theory is making predictions about the chances of occurrence of events based on a set of assumptions about the underlying probability processes.
- Outcome: An outcome is a subset of a probability space. Experimental outcomes are not uniquely determined from the description of an experiment, and must be agreed upon to avoid ambiguity (Papoulis 1984, pp. 24-25).
- Stability: The robustness of a given outcome to small changes in initial conditions or small random fluctuations. Chaos is an example of a process which is not stable.

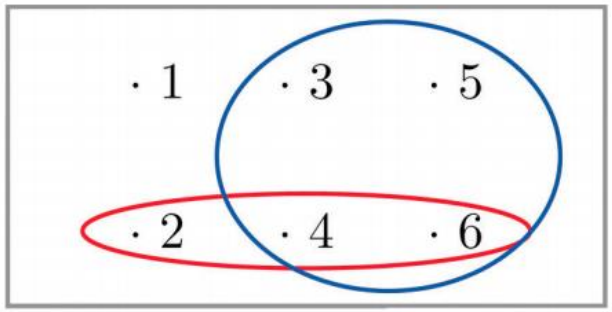
Papoulis, A. *Probability, Random Variables, and Stochastic Processes, 2nd ed.* New York: McGraw-Hill, 1984.

Example

- Construct a sample space for the experiment that consists of rolling a single die. Find the events that correspond to the phrases "an even number is rolled" and "a number greater than two is rolled."

Solution

- The outcomes could be labeled according to the number of dots on the top face of the die. Then the sample space is the set S = {1,2,3,4,5,6}.
- The outcomes that are even are 2, 4, and 6, so the event that corresponds to the phrase "an even number is rolled" is the set {2,4,6}, which it is natural to denote by the letter E. We write E = {2,4,6}.
- Similarly the event that corresponds to the phrase "a number greater than two is rolled" is the set T = {3,4,5,6} , which we have denoted T.



Extrapolation & Standard Error

- Extrapolation: Drawing a conclusion about something beyond data range.
- Extrapolation is needed because of difficulty/impossibility of obtaining data.
- Drawing a conclusion from a biased sample is one form of extrapolation.
- Extrapolation is a common error in applying or interpreting statistics.

- Important aspect of a sampling distribution is standard error (SE).

- Sampling Distribution: Distribution of sample statistics with a mean approximately equal to mean in original distribution and standard deviation known as standard error.
  - Standard Error: Standard deviation of a sampling distribution.

Sampling Error

- Two major sources of error in estimating population parameters: sampling error and bias.
- SE:
  - is random and can sometimes be estimated in advance of study if sufficient information about population is available.
  - addresses how much, on average, sample estimates of a study characteristic or variable differ from sample to sample.
  - essential in describing (research) results.
  - critical in tests of classic statistical significance
  - forms an integral part of generalizing from a sample to larger population because exact population value is typically unknown.

https://methods.sagepub.com/reference/encyc-of-research-design/n401.xml

Sampling Error

- Important statistical construct behind estimating sampling error is *sampling distribution.*
- Simple random samples calculations are basic formulas used for sampling error.
- Larger samples have smaller sampling errors.
- To calculate sampling errors in unequal probability samples are more complicated.
- Calculating sampling errors for multiple-stage designs is more complicated than those for a single-stage design.
- If the sampling distribution approximates a normal curve, by definition 95% of the sample estimates will be within two standard errors on either side of the estimated population parameter.

Sampling Bias

- Bias: systematic favoring of certain outcomes.
- Sampling Bias: Systematic favoring of certain outcomes due to the methods employed to obtain the sample.
- Example: Weight Loss Study Volunteers: A medical research center is testing a new weight loss treatment. They advertise on a social media site that they are looking for volunteers to participate. There is sampling bias because the sample will be limited to people who use the social media site where they advertised. The individuals who choose to participate may be different from the overall population.
- For example, volunteers may be individuals who are already actively trying to lose weight. This is not a representative sample because the sample may have characteristics that are different from the population of interest.

https://online.stat.psu.edu/stat200/lesson/1/1.2/1.2.1

Statistics and Parameters

	Population Parameter	Sample Statistic
Mean	μ	$\bar{x}$
Difference in two means	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$
Proportion	p	$\hat{p}$
Difference in two proportions	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$
Correlation	ρ	r
Slope (simple linear regression)	β	b

https://online.stat.psu.edu/stat200/lesson/4

Estimation

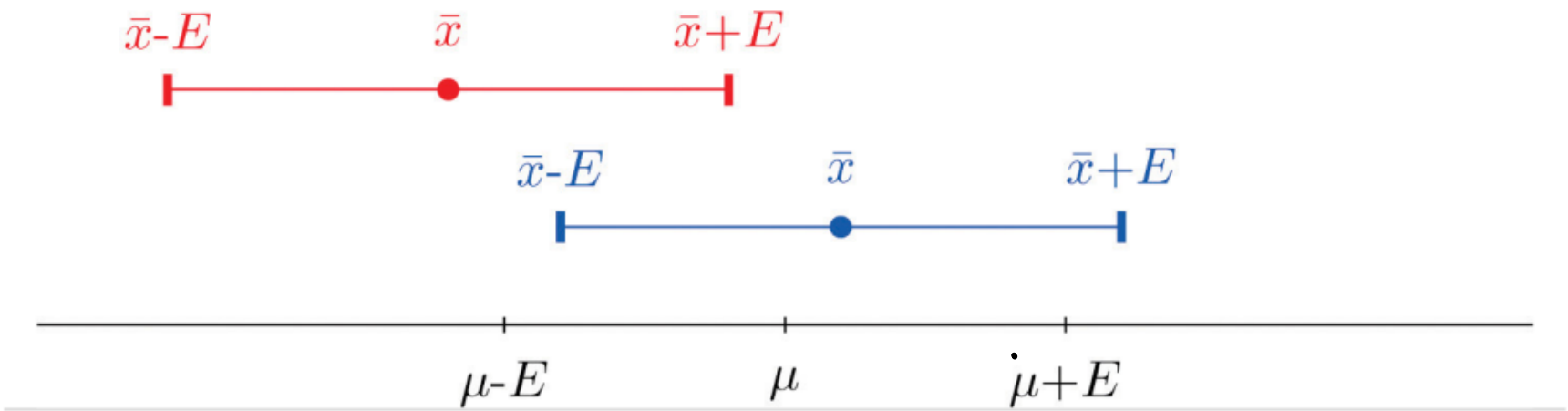
- Estimating a population parameter by a single number is called point estimation.
- Statistic $\bar{x}$ is a point estimate of the parameter μ.
  - Reliability Issue: No indication of how reliable the estimate is.
- E, the number added to and subtracted from the point estimate to produce the interval estimate.
- An interval with endpoints $\bar{x} \pm E$, computed from the sample data in such a way that a specified proportion of all intervals constructed by this process will contain the parameter of interest.

Estimation

- Interval estimates state how much a point estimate is likely to vary by chance.
- Common interval estimate is confidence level, calculated as (1 – a). where a = significance.
- Confidence Interval (CI): is the range of values that we believe will have a specified chance of containing the unknown population parameter.
- CI tells most likely range of the unknown population mean.
- Confidence Level: usually 95%.

Estimation

- Let 95%, of all the intervals constructed from sample data by means of this formula contain the unknown parameter μ. Such an interval is called a 95% confidence interval for μ.
- 95% of the values of $\overline{X}$ lie in the interval [μ – E, μ + E]



More in Detailed

The Central Limit Theorem says that, for large samples (samples of size $n \geq 30$), when viewed as a random variable the sample mean $\overline{X}$ is normally distributed with mean $\mu_{\overline{X}} = \mu$ and standard deviation $\sigma_{\overline{X}} = \sigma \,/\, \sqrt{n}$. The Empirical Rule says that we must go about two standard deviations from the mean to capture 95% of the values of $\overline{X}$ generated by sample after sample. A more precise distance based on the normality of $\overline{X}$ is 1.960 standard deviations, which is $E = 1.960\sigma \,/\, \sqrt{n}$.

Estimation

- Identify the level of confidence in terms of the area α in the two tails of the distribution of $\bar{X}$ when the middle part specified by the level of confidence is taken out.

Large Sample $100(1-\alpha)\%$ Confidence Interval for a Population Mean

If σ is known: $\bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$

If σ is unknown: $\bar{x} \pm z_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)$

A sample is considered large when $n \geq 30$.

Beginning Statistics By Douglas S. Shafer and Zhiyi Zhang.

Large Sample Tests for a Population Mean

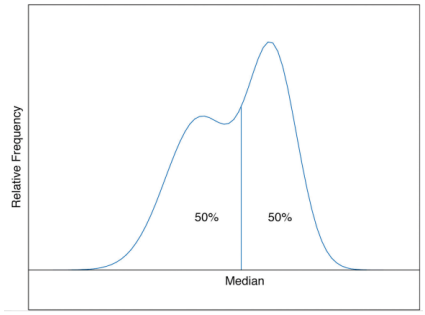
- Central Limit Theorem states that $\bar{X}$ is approximately normally distributed and has mean $\mu_{\bar{X}} = \mu$ and standard deviation $\sigma_{\bar{X}} = \sigma/\sqrt{n}$, where μ and σ are the mean and the standard deviation of the population. This implies that the statistic: $(x - \mu)/(\sigma/\sqrt{n})$ has the standard normal distribution.
- If we do not know σ, then replace it by sample standard deviation s.

$$\text{If } \sigma \text{ is known:} \quad Z = \frac{\bar{x} - \mu_0}{\sigma \big/ \sqrt{n}}$$

$$\text{If } \sigma \text{ is unknown:} \quad Z = \frac{\bar{x} - \mu_0}{s \big/ \sqrt{n}}$$

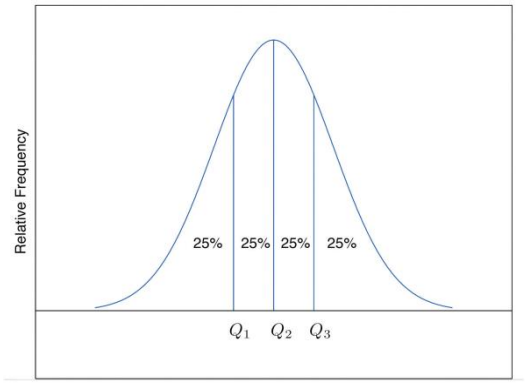
Measures of Central Location

- Sample mean of a set of n sample data is the number x defined by the formula $\bar{x} = \Sigma x/n$
- The population mean of a set of N population data is the number μ defined $\mu = \Sigma x/N$
- Sample median $\tilde{x}$ of a set of sample data for which there are an odd number of measurements is the middle measurement when the data are arranged in numerical order.
- The sample median $\tilde{x}$ of a set of sample data for which there are an even number of measurements is the mean of the two middle measurements when the data are arranged in numerical order.



Quartile

- For any data set: The second quartile Q_2 of the data set is its median.
- Define two subsets:
  - Lower set: all observations that are strictly less than Q_2;
  - Upper set: all observations that are strictly greater than Q_2.
- First quartile Q_1 of the data set is the median of the lower set.
- Third quartile Q_3 of the data set is the median of the upper set.
- Interquartile range (IQR) is the quantity $IQR = Q_3 - Q_1$



Mean/expected value

- The mean of a discrete random variable X is the number

$$\mu = E(X) = \Sigma x\, P(x)$$

- The mean of a random variable may be interpreted as the average of the values assumed by the random variable in repeated trials of the experiment.
- Find the mean of the discrete random variable X whose probability distribution is

x	-2	1	2	3.5
P(x)	0.21	0.34	0.24	0.21

- Soln: The formula in the definition gives $\mu = \Sigma x\, P(x)$

$$= (-2)x0.21+(1)x0.34+(2)x0.24+(3.5)x0.21$$

$$= 1.135$$

Variance and standard deviation

- The variance (σ^2) of a discrete random variable X is the number
- $\sigma^2 = \Sigma(x-\mu)^2 P(x)$
- which by algebra is equivalent to the formula: $\sigma^2 = [\Sigma x^2 P(x)] - \mu^2$
- The standard deviation (σ) of a discrete random variable X is the square root of its variance:
- $\sigma = \sqrt{\Sigma(x-\mu)^2 P(x)}$

  $= \sqrt{[\Sigma x^2 P(x)] - \mu^2}$

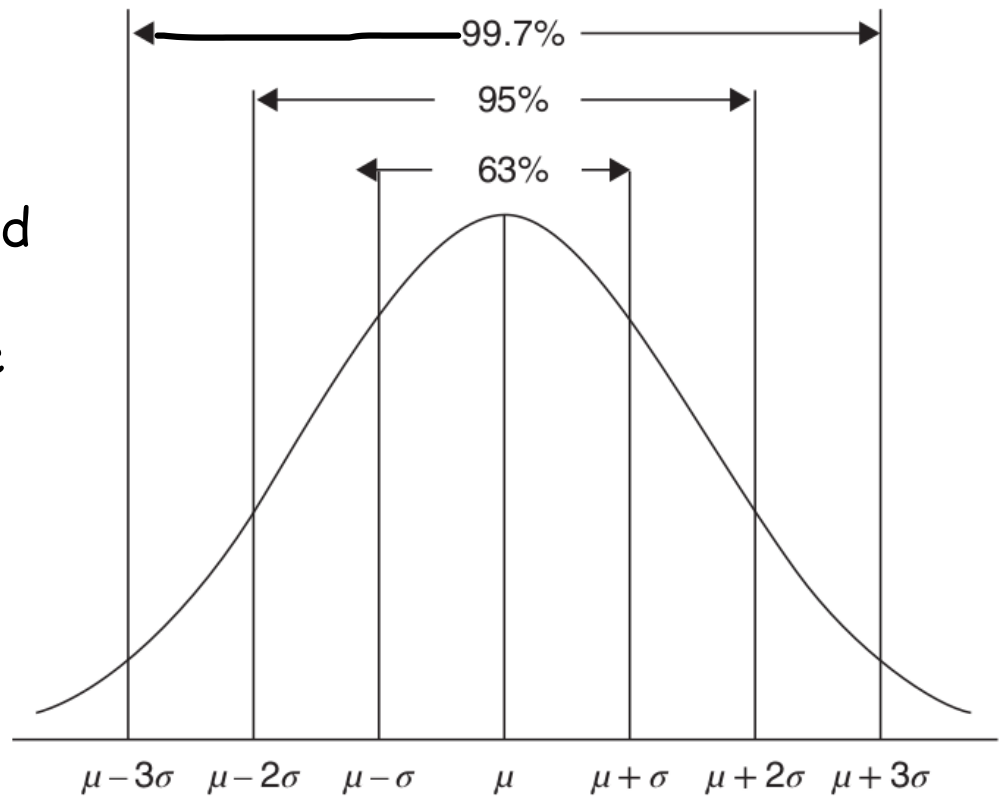
Empirical Rule

- If data distribution is approximately bell-shaped, then empirical rule, can be used to compute the percentage of data that will fall within k standard deviations from the mean (k = 1, 2, 3).
- Rule:
- About 68% of the data will fall within one SD of the mean i.e., between $\mu-1\sigma$ and $\mu+1\sigma$.
- About 95% of the data will fall within two SDs of the mean i.e. between $\mu - 2\sigma$ and $\mu + 2\sigma$.
- About 99.7% of the data will fall within three SDs of the mean i.e. between $\mu-3\sigma$ and $\mu+3\sigma$.

Gupta, Bhisham C., and Irwin Guttman. *Statistics and probability with applications for engineers and scientists*. John Wiley & Sons, 2014.

Empirical Rule

- Where μ and σ are unknown, the empirical rule is of the same form, but μ is replaced by $\overline{X}$ and σ replaced by S.
- If the population data do NOT have bell-shaped distribution, then use Chebyshev's inequality,



Gupta, Bhisham C., and Irwin Guttman. *Statistics and probability with applications for engineers and scientists*. John Wiley & Sons, 2014.

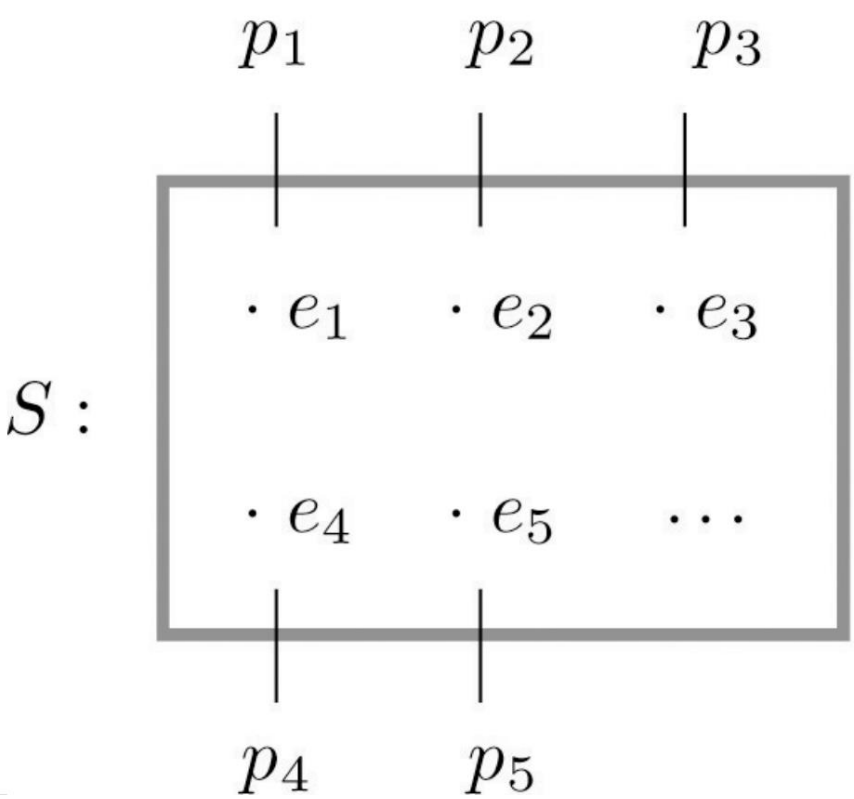
Outline

- Statistical: Very Basic Concepts
- Probability: Very Basic Concepts

Probability

- The probability of an outcome (e) in a sample space (S) is a number p between 0 and 1 that measures the likelihood that (e) will occur on a single trial of the corresponding random experiment.
- If p = 0, the outcome (e) being impossible
- If p = 1, the outcome (e) being certain.
- Every valid probability must be a number between 0 and 1 (or a percent between 0% and 100%).
- The probability of an event (A) is the sum of the probabilities of the individual outcomes of which it is composed. It is denoted P (A) .
- If an event E is $E = \{e_1, e_2, \ldots, e_k\}$, then $P(E) = P(e_1) + P(e_2) + \cdots + P(e_k)$

Sample Spaces and Probability



$$A: \quad \{e_1, e_2\}$$
$$B: \quad \{e_2, e_3, e_4\}$$

$$P(A) = p_1 + p_2$$
$$P(B) = p_2 + p_3 + p_4$$

Example

- A die is called "balanced" or "fair" if each side is equally likely to land on top. Assign a probability to each outcome in the sample space for the experiment that consists of tossing a single fair die. Find the probabilities of the events E: "an even number is rolled" and T: "a number greater than two is rolled."

Solution

- With outcomes labeled according to the number of dots on the top face of the die, the sample space is the set $S = \{1,2,3,4,5,6\}$.
- Since there are six equally likely outcomes, which must add up to 1, each is assigned probability 1/6.
- Since $E = \{2,4,6\}$,

$$P(E) = 1/6 + 1/6 + 1/6$$
$$= 3/6$$
$$= 1/2.$$

- Since $T = \{3,4,5,6\}$,

$$P(T) = 4/6$$
$$= 2/3.$$

Probability Function

- Probability mass functions (pmf) apply to discrete outcomes.
- Pmf list of all possible outcomes for a given experiment and the probabilities for each outcome.

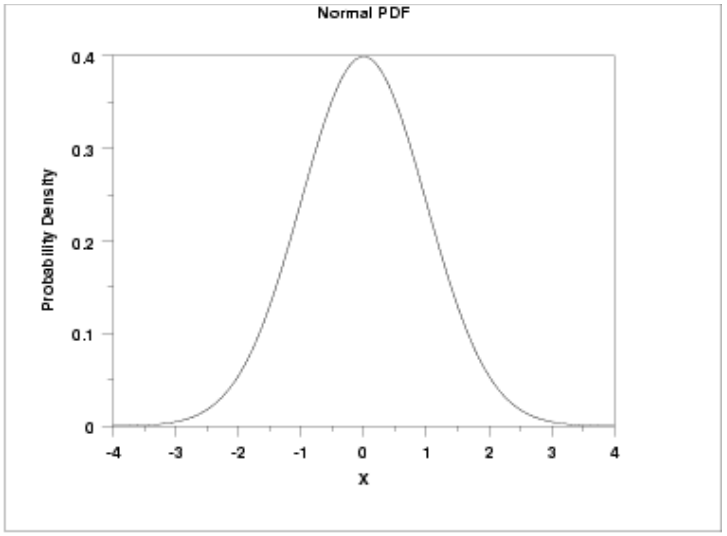
$$f(x) = \begin{cases} 0.25 & \text{if} \quad x = 1 \\ 0.35 & \text{if} \quad x = 2 \\ 0.40 & \text{if} \quad x = 3 \end{cases}$$

where f(x) is the probability that random variable X takes on value x, with f(x)=0 implied for all other x values.

- Probability density function (pdf) represented as "f(x)".
- PDF does not represent probabilities directly but from which the probability that the outcome falls in a certain range can be calculated using integration from calculus.

Probability Density Function

- For continuous function, pdf is the probability that the variate has the value x.
- For continuous distributions, the probability at a single point is zero.
- Expressed as: Integral between two points: $\int_a^b f(x)dx=P[a\leq X\leq b]$
- Discrete distribution: PDF is the probability that the variate takes the value x. $f(x)=Pr[X=x]$.

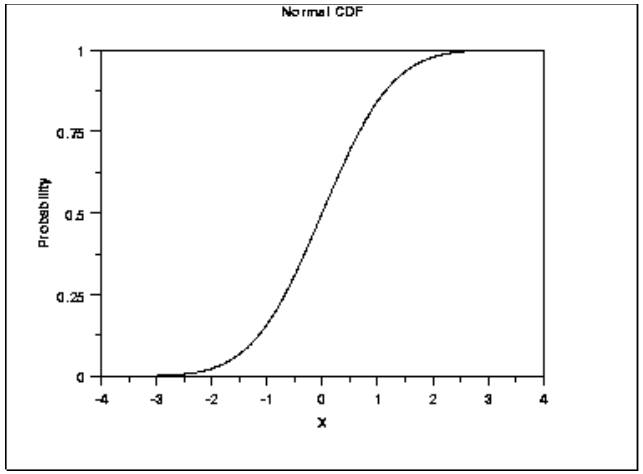


normal PDF

https://www.itl.nist.gov/div898/handbook/eda/section3/eda362.htm

Cumulative Density Function

- CDF is the probability that the variable takes a value less than or equal to x. That is $F(x)=P[X \leq x]=a$
- For a discrete distribution, CDF can be expressed as $F(x)=\sum x_{i=0} f(i)$



Normal CDF

https://www.itl.nist.gov/div898/handbook/eda/section3/eda362.htm

Bonus: Definitions

- ## Cumulative Distribution Functions

Definition. The **cumulative distribution function** ("**c.d.f.**") of a continuous random variable X is defined as:

$$F(x) = \int_{-\infty}^{x} f(t)dt$$

for $-\infty < x < \infty$.

- ## Probability Distribution Functions

Definition. The **probability density function** ("**p.d.f.**") of a continuous random variable X with support S is an integrable function $f(x)$ satisfying the following:

(1) $f(x)$ is positive everywhere in the support S, that is, $f(x) > 0$, for all x in S

(2) The area under the curve $f(x)$ in the support S is 1, that is:

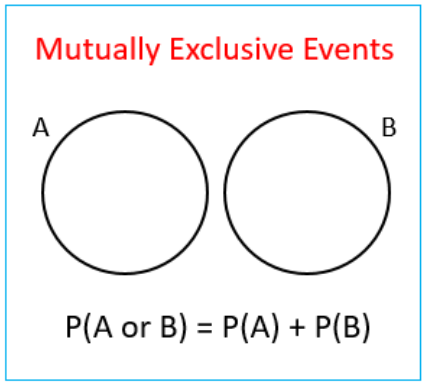
$$\int_S f(x)dx = 1$$

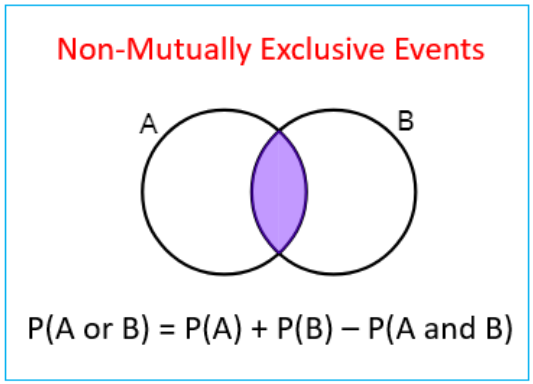
(3) If $f(x)$ is the p.d.f. of x, then the probability that x belongs to A, where A is some interval, is given by the integral of $f(x)$ over that interval, that is:

$$P(X \in A) = \int_A f(x)dx$$

Definitions

- A prior probability is an initial probability value originally obtained before any additional information is obtained.
- A posterior probability is a probability value that has been revised by using additional information that is later obtained.
- Mutually Exclusive Events



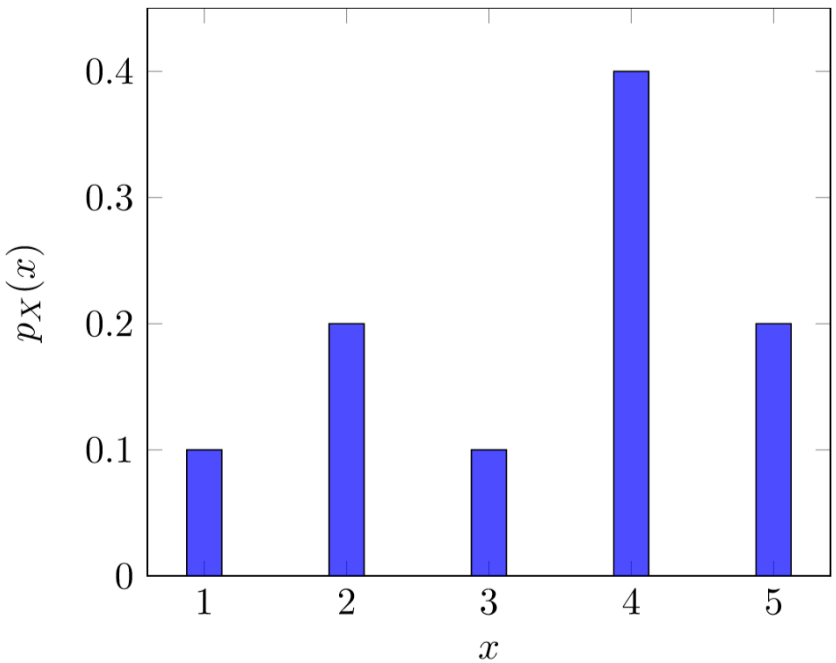


Exercise

- Discrete random variable: Side figure shows the probability mass function of a discrete random variable X (check that it adds up to one). To compute the probability of X belonging to different sets you must apply

$$\mathrm{P}\left(X \in S\right) = \sum_{x \in S} p_X\left(x\right).$$

- Now calculate P(X ∈{1,4}) and P(X > 3).

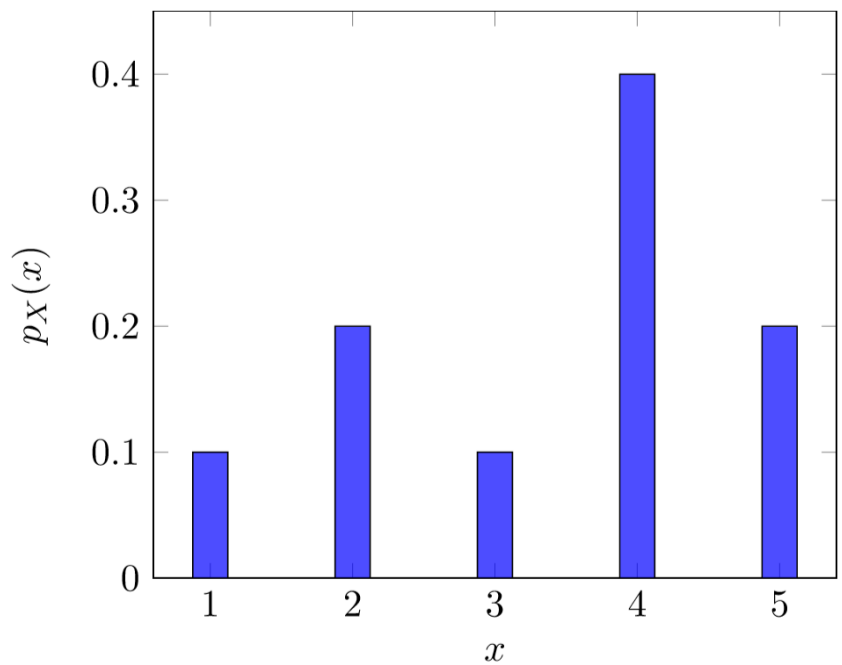


Solution

- Discrete random variable: Side figure shows the probability mass function of a discrete random variable X (check that it adds up to one). To compute the probability of X belonging to different sets you must apply

$$\mathrm{P}\left(X \in S\right) = \sum_{x \in S} p_X\left(x\right).$$

- Now calculate P(X ∈{1,4}) and P(X > 3).



P(X ∈{1,4}) = pX (1) + pX (4) = 0.5
P(X > 3) = pX (4) + pX (5) = 0.6.

Exercise

- Which of the following are probability functions?

a. $f(x)=.25$ for $x=9,10,11,12$

b. $f(x)= (3-x)/2$ for $x=1,2,3,4$

c. $f(x)= (x^2+x+1)/25$ for $x=0,1,2,3$

a. $f(x)=.25$ for $x=9,10,11,12$

x	f(x)
9	.25
10	.25
11	.25
12	.25

1.0 … Yes, probability function!

Solution

- Which of the following are probability functions?

b. $f(x)= (3-x)/2$ for $x=1,2,3,4$

x	f(x)
1	(3-1)/2=1.0
2	(3-2)/2=.5
3	(3-3)/2=0
4	(3-4)/2=-.5

Though this sums to 1, you can't have a negative probability; therefore, it's not a probability function.

Solution

- Which of the following are probability functions?

C. f(x)= (x2+x+1)/25 for x=0,1,2,3

x	f(x)
0	1/25
1	3/25
2	7/25
3	13/25

24/25

Doesn't sum to 1. Thus, it's not a probability function.

Exercise

- The number of ships to arrive at a harbor on any given day is a random variable represented by x. The probability distribution for x is:

x	10	11	12	13	14
P(x)	.4	.2	.2	.1	.1

- Find the probability that on a given day:

a. Exactly 14 ships arrive $p(x=14)= .1$

b. At least 12 ships arrive

c. At most 11 ships arrive

Solution

- The number of ships to arrive at a harbor on any given day is a random variable represented by x. The probability distribution for x is:

x	10	11	12	13	14
P(x)	.4	.2	.2	.1	.1

- Find the probability that on a given day:

a. Exactly 14 ships arrive $p(x=14)= .1$

b. At least 12 ships arrive $p(x>=12)= (.2 + .1 +.1) = .4$

c. At most 11 ships arrive $p(x\leq 11)= (.4 +.2) = .6$

Complements

- Complements: The complement of an event A in a sample space S, denoted A^c , is the collection of all outcomes in S that are not elements of the set A. It corresponds to negating any description in words of the event A.
- Two events connected with the experiment of rolling a single die are E: "the number rolled is even" and T: "the number rolled is greater than two." Find the complement of each.
- <u>Solution</u>: In the sample space S={1,2,3,4,5,6} the corresponding sets of outcomes are E={2,4,6} and T={3,4,5,6}. The complements are E^c={1,3,5} and T^c={1,2}.
- "the number rolled is even"==> "the number rolled is odd" and "the number rolled is greater than two" ==>"the number rolled is less than three.".

Exercise: Complements (1/2)

- Find the probability that at least one heads will appear in five tosses of a fair coin
- <u>Solution</u>: Identify outcomes by lists of five hs and ts, such as tthtt and hhttt. There are two choices for the first toss.
- For each of these there are two choices for the second toss, hence $2 \times 2 = 4$ outcomes for two tosses.
- For each of these four outcomes, there are two possibilities for the third toss, hence $4 \times 2 = 8$ outcomes for three tosses.
- Similarly, there are $8 \times 2 = 16$ outcomes for four tosses and finally $16 \times 2 = 32$ outcomes for five tosses.

Exercise: Complements (2/2)

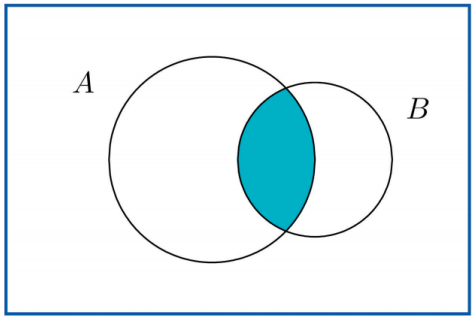
- Let O denote the event "at least one heads."
- There are many ways to obtain at least one heads, but only one way to fail to do so: all tails.
- Thus although it is difficult to list all the outcomes that form O, it is easy to write O^c = {ttttt}.
- Since there are 32 equally likely outcomes, each has probability 1/32, so $P(O^c) = 1/32$,

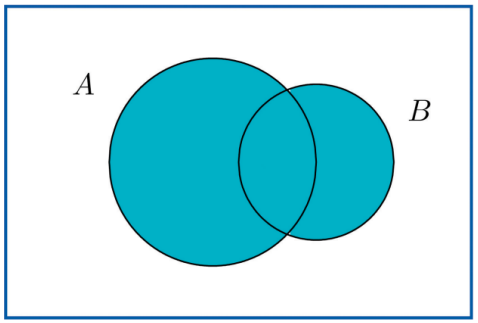
hence $P(O) = 1 - 1/32 \approx 0.97$ or about a 97% chance.

Intersection and Union

- Probability Rule for Complements: $P(A^c) = 1 - P(A)$.
  - Useful when finding the probability of an event directly is difficult.
- Intersection of Events: The intersection of events A and B, denoted $A \cap B$, is the collection of all outcomes that are elements of both sets A and B. It corresponds to combining descriptions of the two events using the word "and."
- Union of Events: The union of events A and B, denoted $A \cup B$, is the collection of all outcomes that are elements of one or the other of the sets A and B, or of both. It corresponds to combining descriptions of the two events using the word "or."

Intersection of Events





Union of Events

Example on intersection

- In the experiment of rolling a single die, find the intersection $E \cap T$ of the events E: "the number rolled is even" and T: "the number rolled is greater than two."
- Soln: The sample space is $S = \{1,2,3,4,5,6\}$.

Since the outcomes that are common to $E = \{2,4,6\}$ and $T = \{3,4,5,6\}$ are 4 and 6, $E \cap T = \{4,6\}$.

In words the intersection is described by "the number rolled is even and is greater than two." The only numbers between one and six that are both even and greater than two are four and six, corresponding to $E \cap T$ given above.

Example on Union

- Additive Rule of Probability: Useful for calculating the probability of $A \cup B$. Formula: $P(A \cup B)=P(A)+P(B)-P(A \cap B)$.
- In the experiment of rolling a single die, find the union of the events E: "the number rolled is even" and T: "the number rolled is greater than two."
- <u>Solution</u>: Since the outcomes that are in either $E = \{2,4,6\}$ or $T = \{3,4,5,6\}$ (or both) are 2, 3, 4, 5, and 6, $E \cup T = \{2,3,4,5,6\}$.
- Note that an outcome such as 4 that is in both sets is still listed only once.
- In words the union is described by "the number rolled is even or is greater than two." Every number between one and six except the number one is either even or is greater than two, corresponding to $E \cup T$ given above.

Exercise

- A tutoring service specializes in preparing adults for high school equivalence tests. Among all the students seeking help from the service, 63% need help in mathematics, 34% need help in English, and 27% need help in both mathematics and English. What is the percentage of students who need help in either mathematics or English?

Solution

- Imagine selecting a student at random, that is, in such a way that every student has the same chance of being selected. Let M denote the event "the student needs help in mathematics" and let E denote the event "the student needs help in English."
- The information given is that $P(M) = 0.63$,
- $P(E) = 0.34$, and $P(M \cap E) = 0.27$.
- The Additive Rule of Probability gives $P(M \cup E) = P(M) + P(E) - P(M \cap E)$

$$= 0.63 + 0.34 - 0.27$$

$$= 0.70$$

Example on intersection

- A single die is rolled.
  - a. Suppose the die is fair. Find the probability that the number rolled is both even and greater than two.
  - b. Suppose the die has been "loaded" so that P (1) = 1/12, P(6)=3/12, and the remaining four outcomes are equally likely with one another. Now find the probability that the number rolled is both even and greater than two.

Solution on intersection

- In both cases the sample space is $S = \{1,2,3,4,5,6\}$ and the event in question is the intersection $E \cap T = \{4,6\}$.
- a. Since the die is fair, all outcomes are equally likely, so by counting we have $P(E \cap T) = 2/6$.
- b. The information on the probabilities of the six outcomes that we have so far is

Outcome	1	2	3	4	5	6
Probablity	1/12	p	p	p	p	3/12

Since $P(1) + P(6) = 4/12 = 1/3$ and the probabilities of all six outcomes add up to 1,

$P(2)+P(3)+P(4)+P(5)=1-1/3=2/3$.

Thus $4p = 2/3$, so $p = 1/6$. In particular, $P(4) = 1/6$.

Therefore, $P(E \cap T) = P(4) + P(6) = 1/6 + 3/12 = 5/12$.

Mutually Exclusive

- Events A and B are mutually exclusive* if they have no elements in common.
- Probability Rule for Mutually Exclusive Events:
  - Events A and B are mutually exclusive if and only if, $P(A \cap B) = 0$
- Any event A and its complement A^c are mutually exclusive, but A and B can be mutually exclusive without being complements.
- Example: In the experiment of rolling a single die, find three choices for an event A so that the events A and E: "the number rolled is even" are mutually exclusive.
- <u>Solution</u>: Since $E = \{2,4,6\}$ and we want A to have no elements in common with E, any event that does not contain any even number will do. Three choices are $\{1,3,5\}$ (the complement E^c, the odds), $\{1,3\}$, and $\{5\}$.

*Events that cannot both occur at once.

Conditional Probability

- The conditional probability of A given B, denoted $P(A|B)$, is the probability that event A has occurred in a trial of a random experiment for which it is known that event B has occurred.
- Conditional Probability $P(A|B) = P(A \cap B)/P(B)$
- Example: A fair die is rolled.

1. Find the probability that the number rolled is a five, given that it is odd.
2. Find the probability that the number rolled is odd, given that it is a five.

Solution#1

- The sample space for this experiment is the set $S = \{1,2,3,4,5,6\}$ consisting of six equally likely outcomes.
- Let F denote the event "a five is rolled" and let O denote the event "an odd number is rolled," so that $F = \{5\}$ and $O = \{1,3,5\}$
- a: Following the formula: $P(F|O) = P(F \cap O)/P(O)$,
- Since $F \cap O = \{5\} \cap \{1,3,5\} = \{5\}$, $P(F \cap O) = 1/6$.
- Since $O = \{1,3,5\}$, $P(O) = 3/6$.
- Finally, $P(F|O) = P(F \cap O)/P(O) = \{1/6\}/\{3/6\} = 1/3$

Solution#2

- 2:This is the same problem, but with the roles of F and O reversed.
- Now TRY by yourself...

Solution#2

- 2:This is the same problem, but with the roles of F and O reversed.

$P(O|F) = P(O \cap F)/P(F)$

Obviously, $P(F) = 1/6$. In part (a) we found that $P(F \cap O) = 1/6$.

Thus $P(O|F) = P(O \cap F)/P(F) = \{1/6\}/\{1/6\} = 1$

Independent Events

- Events A and B are independent, if $P(A \cap B) = P(A) \cdot P(B)$
- Events whose probability of occurring together is the product of their individual probabilities.
- If A and B are not independent, then they are dependent.
- Example: A single fair die is rolled. Let $A = \{3\}$ and $B = \{1,3,5\}$. Are A and B independent?
- Soln: We can compute all three probabilities $P(A)= 1/6$, $P(B) =1/2$, and $P(A \cap B)=P(3)= 1/6$.
- Since the product $P(A) \cdot P(B) = (1/6).(1/2)=1/12$ is not the same number as $P(A \cap B)=1/6$, the events A and B are not independent.

Random Variables

- A random variable is a numerical quantity that is generated by a random experiment.
- Example:
  - Roll two fair dice
  - Number of tosses until the coin lands heads
  - Possible Values of x in 1, 2, 3,4, …
- A random variable is called discrete if it has either a finite or a countable number of possible values.
- A random variable is called continuous if its possible values contain a whole interval of numbers.

Probability Distributions

- The probability distribution of a discrete random variable X is a list of each possible value of X together with the probability that X takes that value in one trial of the experiment.
- The probabilities in the probability distribution of a random variable X must satisfy these conditions:
- Each probability P(x) must be between 0 and 1: $0 \leq P(x) \leq 1$.
- Sum of all the probabilities is 1: $\Sigma P(x) = 1$

Example (1/2)

- A fair coin is tossed twice. Let X be the number of heads that are observed.
- a. Construct the probability distribution of X.
- b. Find the probability that at least one head is observed.
- Soln:
- a. The possible values that X can take are 0, 1, and 2. Each of these numbers corresponds to an event in the sample space S = {hh, ht, th, tt} of equally likely outcomes for this experiment: X = 0 to {tt}, X = 1 to {ht, th}, and X = 2 to {hh}.
- The probability of each of these events, hence of the corresponding value of X, can be found simply by counting

Example (2/2)

- This table is the probability distribution of X.

x	0	1	2
P(x)	0.25	0.5	0.25

- b. "At least one head" is the event $X \geq 1$, which is the union of the mutually exclusive events $X = 1$ and $X = 2$. Thus
- $P(X \geq 1) = P(1) + P(2) = 0.50 + 0.25 = 0.75$

More Examples

Assignment#1

- A random experiment consists of tossing two coins.
- a. Construct a sample space for the situation that the coins are indistinguishable, such as two brand new pennies.
- b. Construct a sample space for the situation that the coins are distinguishable, such as one a penny and the other a nickel.

Solution#1

- a. After the coins are tossed one sees either two heads, which could be labeled 2h, two tails, which could be labeled 2t, or coins that differ, which could be labeled d. Thus a sample space is S = {2h, 2t, d}.
- b. Since we can tell the coins apart, there are now two ways for the coins to differ:
  - the penny heads and the nickel tails,
  - or the penny tails and the nickel heads.
- We can label each outcome as a pair of letters, the first of which indicates how the penny landed and the second of which indicates how the nickel landed. A sample space is then S' = {hh, ht, th, tt} .

Assignment#2

- The student body in the high school may be broken down into ten categories as follows: 25% white male, 26% white female, 12% black male, 15% black female, 6% Hispanic male, 5% Hispanic female, 3% Asian male, 3% Asian female, 1% male of other minorities combined, and 4% female of other minorities combined. A student is randomly selected from this high school. Find the probabilities of the following events:
  1. B: the student is black,
  2. MF: the student is minority female,
  3. FN: the student is female and is not black.

Solution#2

- Now the sample space is S = {wm, bm, hm, am, om, wf, bf, hf, af, of}. The information given in the example can be summarized in the following table, called a two-way contingency table:

Gender	Race/Ethnicity				
	White	Black	Hispanic	Asian	Others
Male	0.25	0.12	0.06	0.03	0.01
Female	0.26	0.15	0.05	0.03	0.04

- a: Since B = {bm, bf} , P(B) = P(bm) + P(bf) = 0.12 + 0.15 = 0.27.
- b: Since MF = {bf, hf, af, of}, P(M)=P(bf)+P(hf)+P(af)+P(of)=0.15+0.05+0.03+0.04=0.27
- c. Since FN = {wf, hf, af, of}, P(FN)=P(wf)+P(hf)+P(af)+P(of)=0.26+0.05+0.03+0.04=0.38

Assignment#3

- Two fair dice are thrown. Find the probabilities of the following events:
- a. both dice show a four
- b. at least one die shows a four (Hints: Additive Rule of Probability)

Solution#3

- As was the case with tossing two identical coins, actual experience dictates that for the sample space to have equally likely outcomes we should list outcomes as if we could distinguish the two dice.
- We could imagine that one of them is red and the other is green. Then any outcome can be labeled as a pair of numbers as in the following display, where the first number in the pair is the number of dots on the top face of the green die and the second number in the pair is the number of dots on the top face of the red die.

11	12	13	14	15	16
21	22	23	24	25	26
31	32	33	34	35	36
41	42	43	44	45	46
51	52	53	54	55	56
61	62	63	64	65	66

Solution#3

- a. There are 36 equally likely outcomes, of which exactly one corresponds to two fours, so the probability of a pair of fours is 1/36.
- b. From the table we can see that there are 11 pairs that correspond to the event in question: the six pairs in the fourth row (the green die shows a four) plus the additional five pairs other than the pair 44, already counted, in the fourth column (the red die is four), so the answer is 11/36. To see how the formula gives the same number, let A G denote the event that the green die is a four and let A R denote the event that the red die is a four.
- Then clearly by counting we get $P(A_G) = 6/36$ and $P(A_R) = 6/36$. Since $A_G \cap A_R = \{44\}$, $P(A_G \cap A_R) = 1/36$; this is the computation in part (a), of course. Thus by the Additive Rule of Probability, $P(A_G \cup A_R) = P(A_G) + P(A_R) - P(A_G - A_R) = 6/36+6/36-1/36=11/36$.

Assignment#4

- Volunteers for a disaster relief effort were classified according to both specialty (C: construction, E: education, M: medicine) and language ability (S: speaks a single language fluently, T: speaks two or more languages fluently). The results are shown in the following two-way classification table. The first row of numbers means that 12 volunteers whose specialty is construction speak a single language fluently, and 1 volunteer whose specialty is construction speaks at least two languages fluently. Similarly for the other two rows. A volunteer is selected at random, meaning that each one has an equal chance of being chosen. Find the probability that:

1. his specialty is medicine and he speaks two or more languages;
2. either his specialty is medicine or he speaks two or more languages;
3. his specialty is something other than medicine.

Speciality	Language Ability	
	S	T
C	12	1
E	4	3
M	6	2

Solution: 4.1

Speciality	Language Ability		
	S	T	Total
C	12	1	13
E	4	3	7
M	6	2	8
Total	22	6	28

- The probability sought is $P(M \cap T)$. The table shows that there are 2 such people, out of 28 in all, hence $P(M \cap T) = 2/28 \approx 0.07$ or about a 7% chance.

Solution: 4.2

- The probability sought is $P(M \cup T)$.
- The third row total and the grand total in the sample give $P(M) = 8/28$.
- The second column total and the grand total give $P(T) = 6/28$.
- Thus using the result from part (a),

$P(M \cup T) = P(M) + P(T) - P(M \cap T) = 8/28 + 6/28 - 2/28 = 12/28$ or about a 43% chance.

Solution: 4.3

This probability can be computed in two ways. Since the event of interest can be viewed as the event $C \cup E$ and the events C and E are mutually exclusive, the answer is, using the first two row totals,

$P(C \cup E) = P(C) + P(E) - P(C \cap E) = 13/28 + 7/28 - 0/28 = 20/28 = 0.71$

On the other hand, the event of interest can be thought of as the complement M^c of M, hence using the value of $P(M)$ computed in part (b), $P(M^c) = 1 - P(M) = 1 - 8/28 = 20/28 \approx 0.71$ as before.

The probability of an event that is a complement or union of events of known probability can be computed using formulas.